DISCRETE STRUCTURES

CMSC250

Chapter 16: Recursion and Structural Induction

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Recursive objects, strings, trees, and structural proofs



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1 Recursively defined objects

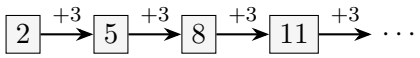
A recursive definition describes a collection by giving **base objects** and **rules for building new objects** from previously constructed ones. Unlike a sequence indexed by integers, the objects may be strings, expressions, lists, or trees.

Key idea. Two parts of a recursive definition: (1) base cases specify objects already in the set; (2) constructor rules specify how to form new objects. Unless stated otherwise, nothing else belongs to the set.

Worked example: a recursively defined set of integers

Question. Let S be the smallest set such that $2 \in S$ and whenever $x \in S$, then $x + 3 \in S$. List its first five elements.

Solution. Begin with the base element 2. Apply the constructor repeatedly: $2, 5, 8, 11, 14, \ldots$. In fact $S = \{2 + 3k : k \in \mathbb{Z},\ k \geq 0\}$. The phrase "smallest set" prevents unrelated elements from appearing.



2 Structural induction: the main principle

Structural induction proves a property of every object generated by a recursive definition. Prove it for each base object. Then, for each constructor, assume the property for the immediate component objects and prove it for the newly built object.

1. State the property $P(x)$ for every recursively generated object x.
2. Check each base object.
3. Consider each constructor and assume P holds for its inputs.
4. Prove P for its output; conclude for all generated objects. ■

3 Structural induction on integer sets

Worked example: every generated number has a certain remainder

Question. Let S be defined by $2 \in S$ and $x \in S \Rightarrow x + 3 \in S$. Prove every $x \in S$ satisfies $x \equiv 2 \pmod 3$.

Proof. Define $P(x)$ to mean $x = 3t + 2$ for some integer t. **Base:** $2 = 3 \cdot 0 + 2$, so $P(2)$ holds. **Constructor step:** Let $x \in S$ be any previously generated element, and assume $x = 3t + 2$ for an integer t. The constructor creates $x + 3 = 3(t + 1) + 2$. Since $t + 1 \in \mathbb{Z}$, the new element satisfies P. By structural induction, every element of S is congruent to 2 modulo 3. ■

Why the proof works. This proves every generated element has the property. It does not by itself prove that *every* integer congruent to 2 modulo 3 belongs to S: negative values such as -1 are excluded.

4 Strings: recursive construction and length

Let B be the smallest collection of strings containing the empty string ε and satisfying: if $w \in B$, then $0w1 \in B$. The first few strings are ε, 01, 0011, 000111, and so on. The length $|w|$ is the number of symbols in w, with $|\varepsilon| = 0$.

Worked example: every recursively constructed string has equal zeros and ones

Question. Prove that every string $w \in B$ contains the same number of 0s and 1s.

Proof. Let $z(w)$ and $o(w)$ denote the number of zeros and ones in w. Define $P(w)$ as $z(w) = o(w)$. **Base:** $z(\varepsilon) = o(\varepsilon) = 0$. **Constructor step:** Assume $P(w)$ for an arbitrary previously constructed $w \in B$. The new string $0w1$ has $z(0w1) = z(w) + 1$ and $o(0w1) = o(w) + 1$. These are equal by the structural IH. Thus $P(0w1)$ holds. Structural induction proves the claim. ■

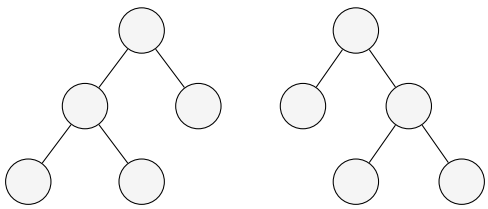
Worked example: every constructed string has even length

Question. Prove $|w|$ is even for all $w \in B$.

Proof. Base: $|\varepsilon| = 0 = 2 \cdot 0$. **Constructor step:** Assume $|w| = 2t$ for some integer $t \geq 0$. Then $|0w1| = |w| + 2 = 2(t + 1)$, which is even. By structural induction, all strings in B have even length. ■

5 Binary trees: a visual recursive definition

A **full binary tree** is defined recursively: (1) a single leaf is a full binary tree; (2) if L and R are full binary trees, then a new root with left subtree L and right subtree R is a full binary tree. Every internal node therefore has exactly two children.



Let $L(T)$ count leaves and $I(T)$ count internal nodes. A single leaf has $L = 1$, $I = 0$. A new root with subtrees A, B has

$$L(T) = L(A) + L(B), \qquad I(T) = I(A) + I(B) + 1.$$

6 The leaf/internal-node identity

Worked example: a full binary tree has one more leaf than internal nodes

Question. Prove $L(T) = I(T) + 1$ for every full binary tree T.

Proof. Let $P(T)$ assert $L(T) = I(T) + 1$. **Base:** If T is a single leaf, $L(T) = 1 = 0 + 1 = I(T) + 1$. **Constructor step:** Let A, B be arbitrary full binary trees and assume $L(A) = I(A) + 1$ and

$L(B) = I(B) + 1$. Construct T with a new root and subtrees A, B. Then

$$\begin{aligned} L(T) &= L(A) + L(B) \\ &= (I(A) + 1) + (I(B) + 1) && \text{structural IH} \\ &= I(A) + I(B) + 2 \\ &= (I(A) + I(B) + 1) + 1 = I(T) + 1. \end{aligned}$$

The property is preserved by the constructor. By structural induction, $L(T) = I(T) + 1$ for every full binary tree. ■

7 Counting edges in a full binary tree

Worked example: twice as many edges as internal nodes

Question. Prove every full binary tree T has $E(T) = 2I(T)$ edges.

Proof. Base: A single leaf has $E = 0 = 2 \cdot 0$. **Constructor step:** Let A, B be full binary trees satisfying $E(A) = 2I(A)$ and $E(B) = 2I(B)$. The new root adds exactly two edges, so

$$E(T) = E(A) + E(B) + 2 = 2I(A) + 2I(B) + 2 = 2(I(A) + I(B) + 1) = 2I(T).$$

By structural induction, the identity holds for every full binary tree. ■

8 Structural recursion: computing rather than proving

A **structurally recursive function** gives an output on base objects and combines outputs for constructor inputs. For a full binary tree,

$$\text{nodes}(\text{leaf}) = 1, \qquad \text{nodes}(\text{node}(A, B)) = 1 + \text{nodes}(A) + \text{nodes}(B).$$

This is a *definition of a computation*, not a proof. Structural induction can then verify a property of that computation.

Worked example: height of a full binary tree

Question. Define height recursively when the height of a single leaf is zero.

Solution. For the base case, $h(\text{leaf}) = 0$. For a root with subtrees A, B, define $h(\text{node}(A, B)) = 1 + \max\{h(A), h(B)\}$. The maximum matters because the longest root-to-leaf path determines height.

9 Induction on numbers versus on structures

Ordinary or strong induction follows an integer parameter n. Structural induction follows the *constructors* that build an object. For trees, the induction hypotheses may be needed for two separate subtrees, even though neither is naturally called "the previous tree."

10 Common mistakes

- **Ignoring a constructor:** every allowed construction rule needs a proof step.
- **Using a property only on the whole tree:** the structural IH applies to the component subtrees.
- **Confusing full and arbitrary binary trees:** the leaf identity above relies on each internal node having exactly two children.
- **Claiming an exact description from one inclusion:** showing every generated object has a property does not automatically show every object with that property is generated.

11 Practice questions with complete solutions

Question. 1. Let S be generated by $1 \in S$ and $x \in S \Rightarrow x + 4 \in S$. Prove all elements of S are odd.

Solution. Base: $1 = 2 \cdot 0 + 1$. Suppose a generated $x = 2t + 1$ for some integer t. Then $x + 4 = 2(t + 2) + 1$, which is odd because $t + 2 \in \mathbb{Z}$. Structural induction proves every element of S is odd. ■**Question.** 2. In the string system B above, prove every nonempty string starts with 0 and ends with 1.

Solution. The base string ε is empty, so the assertion is vacuously true there. For the constructor, any string of the form $0w1$ is nonempty, starts with 0, and ends with 1, regardless of the contents of w. Thus every generated nonempty string has the property. ■**Question.** 3. Prove the total number of nodes $N(T)$ in a full binary tree is odd.

Solution. Base: a leaf has $N = 1$, odd. Suppose $N(A) = 2r + 1$ and $N(B) = 2s + 1$ for integers r, s. The new tree has $N(T) = 1 + N(A) + N(B) = 2(r + s + 1) + 1$, odd. Structural induction proves the claim. ■